

Policy Markets: An AMM for Committed Rebalancing Programs

Working Paper — Draft

Abstract

We introduce Policy Markets, a mechanism where traders commit to rebalancing programs—specified by a drift angle, sensitivity, and size—and receive assured counter-policies at reduced sensitivity. An automated market maker with a sinh-product invariant prices these commitments, with the sensitivity ratio serving as a permanent execution quality locked into the position for its entire life. The underlying asset emerges as the trivial policy at consensus drift with unit sensitivity. The mechanism combines properties of perpetual swaps and cross-currency interest rate swaps, with the funding rate crystallized at entry rather than applied periodically. The hyperbolic geometry of the pricing surface arises endogenously from the equilibrium depth profile of the book.

1 Introduction

Existing derivatives markets separate two logically linked concerns: the expression of a directional view and the execution of that view over time. A trader entering a perpetual swap commits to directional exposure, then faces a stream of unpredictable funding rate payments that represent the ongoing cost of maintaining that exposure. The funding rate is repriced periodically based on market skew—a quantity the trader cannot control or predict at entry.

We propose an alternative in which the full execution cost is determined at entry and embedded in the position as a structural parameter. The traded instrument is a *policy*: a committed rebalancing program specified by three parameters. The market prices these policies against each other and against the underlying asset via an invariant that preserves the pool’s neutrality.

2 The Primitive

Definition 1 (Policy). *A policy is a tuple (φ, w, q) where:*

- $\varphi \in \mathbb{R}$ is the **drift angle**—the trader’s expected drift of the asset price, defining a reference path $r(t; \varphi)$ against which deviations are measured. The drift is signed relative to consensus φ^* ; there is no separate “long/short” designation.
- $w \in \mathbb{R}^+$ is the **sensitivity** (rebalancing intensity), living on a multiplicative scale:
 - $w = 1$: trivial policy; just hold q units. The fixed point of the w -axis.
 - $w > 1$: trend-following; exposure moves more sharply than passive holding. The trader’s weight in the asset amplifies deviations from their drift path.
 - $w < 1$: contrarian; exposure moves less sharply than holding, or opposite. Dampened or reversed response to deviations.

The log convention is natural: $w = 2$ and $w = 1/2$ are mirror-opposite intensities. Conviction is $|\varphi - \varphi^*|$; direction is $\text{sign}(\varphi - \varphi^*)$.

- $q \in \mathbb{R}^+$ is the number of **units**.

Definition 2 (The Asset). The underlying asset (e.g. BTC) is the trivial policy:

$$BTC \equiv (\varphi^*, 1, 1) \quad (1)$$

A commitment at the market's consensus drift φ^* , unit sensitivity, unit size. Holding the asset is agreeing with the market's drift consensus at unit sensitivity—the “do nothing” policy. Every other policy is a deviation from this, priced by the invariant.

3 The Trade

A trader commits $(\varphi_i, w_{\text{in}}, q)$ and receives a counter-policy $(\varphi_i, w_{\text{out}}, q)$ in the opposite direction:

- Same drift angle φ_i on both legs.
- Same units q on both legs.
- Opposite direction.
- **Reduced sensitivity**: the counter-policy is closer to the trivial policy ($w = 1$) than the committed policy. In the trend-following regime ($w > 1$): $w_{\text{out}} < w_{\text{in}}$. In the contrarian regime ($w < 1$): $w_{\text{out}} > w_{\text{in}}$, toward 1. In both cases, the counter-policy has been pulled toward the trivial policy.

Definition 3 (Sensitivity Ratio). The **policy price** is the sensitivity ratio:

$$\rho \in (0, 1] \quad (2)$$

interpreted as value-preservation efficiency: the fraction of committed value that survives execution. The counter-policy sensitivity and net residual are computed in log-space:

$$\boxed{w_{\text{out}} = w_{\text{in}}^\rho, \quad w_{\text{net}} = w_{\text{in}}^{1-\rho}} \quad (3)$$

At $\rho = 1$ (perfect efficiency): $w_{\text{out}} = w_{\text{in}}$, $w_{\text{net}} = 1$ —the counter-policy fully matches the commitment and the residual is the trivial policy. At $\rho \rightarrow 0$: $w_{\text{out}} \rightarrow 1$, $w_{\text{net}} \rightarrow w_{\text{in}}$ —the counter-policy collapses to just holding, and the entire commitment is unmatched.

Remark 1. The log-space convention is structurally motivated. Because w lives on a multiplicative scale with fixed point at 1, the natural interpolation between committed and trivial sensitivity is exponential. The sensitivity ratio acts as an exponent: it places the counter-policy at a well-defined fraction (in log-space) of the way between the committed policy and the trivial policy. This ensures $w = 2$ (trend-following) and $w = 1/2$ (contrarian) are mirror-opposite commitments with the same effective size magnitude.

Remark 2. The sensitivity ratio is not a one-time entry cost. It is pre-determined slippage on all future rebalances: given whatever price the market is at, the effective fill on every future rebalance will be at this locked-in efficiency relative to the clearing quote.

4 The Automated Market Maker

4.1 Pool State and Invariant

The pool maintains two **wing weights** representing cumulative commitment mass above and below consensus:

$$L = \sinh(\Phi_L), \quad S = \sinh(\Phi_S) \quad (4)$$

L is the wing weight above consensus φ^* (commitments with $\varphi > \varphi^*$); S is the wing weight below ($\varphi < \varphi^*$). These are not token reserves—they are aggregate commitment mass in sinh-angle coordinates.

Definition 4 (Invariant).

$$\boxed{L \cdot S = K \iff \sinh(\Phi_L) \cdot \sinh(\Phi_S) = K} \quad (5)$$

K is conserved across trades. All trades move the pool state along this curve. Liquidity events (entries and exits) change K .

The pool state is two-dimensional: equivalently (L, S) , or (K, tilt) where:

$$\text{tilt} = \frac{L - S}{L + S} \quad (6)$$

4.2 The Pool as the Asset

The pool holds both sides of all intermediated rebalancing. At balance ($L = S = \sqrt{K}$), the commitment mass above and below consensus cancel exactly. The pool has zero net exposure to deviations from consensus—it tracks consensus drift perfectly and does not respond to fluctuations around it. Its net sensitivity is always 1.

Proposition 1. *The invariant is the statement: the pool stays at sensitivity 1.*

Argument. The pool's exposure to a deviation ϵ from consensus is $dV/d\epsilon = L - S$. At the consensus frame (where $L = S$), this vanishes. The pool responds to the *level* of the consensus price (sensitivity 1 to the asset) but has zero sensitivity to deviations from consensus. The invariant $L \cdot S = K$ ensures that any trade that skews the pool ($L \neq S$) is accompanied by a consensus shift that restores $L = S$ in the new frame. $\square$

4.3 Consensus and Marginal Quote

$$\varphi^* = \frac{\sum_i w_i \varphi_i}{\sum_i w_i} \quad (7)$$

Consensus is the frame where the pool balances. It moves as commitments enter, exit, or revise. The BTC/USD price at time t is $S(t) = r(t; \varphi^*)$. The marginal quote:

$$P = \frac{S}{L} \quad (8)$$

At balance ($L = S = \sqrt{K}$): $P = 1$. Commitments on either side of consensus trade one-for-one.

5 Trade Mechanics

5.1 Effective Size

A commitment $(\varphi_i, w_{\text{in}}, q)$ maps to a signed effective size:

$$\boxed{\delta = q \cdot \log(w_{\text{in}}) \cdot \sinh(\varphi_i - \varphi^*)} \quad (9)$$

The sign of δ determines which wing absorbs the commitment: $\delta > 0$ pushes into the L -wing (above consensus); $\delta < 0$ pushes into the S -wing (below consensus).

Three structural properties of this formula:

1. At $w = 1$: $\log(1) = 0$, so $\delta = 0$ regardless of drift. The trivial policy has no effective size—it does not interact with the pool.
2. w and $1/w$ produce equal-magnitude, opposite-sign δ for the same drift gap. Trend-following and contrarian at mirror sensitivities are mirror trades.
3. $\sinh(\varphi_i - \varphi^*)$ makes extreme drift convictions exponentially expensive.

5.2 Execution

Without loss of generality, consider $\delta > 0$ (commitment absorbed by the L -wing). The L -wing absorbs $|\delta|$:

$$L' = L + |\delta|, \quad S' = \frac{K}{L + |\delta|} \quad (10)$$

For $\delta < 0$, the roles of L and S are exchanged.

5.3 The Sensitivity Ratio

Definition 5 (Sensitivity Ratio — Derivation). *The sensitivity ratio is the value-preservation efficiency of the trade:*

$$\boxed{\rho = \frac{L}{L + |\delta|}} \quad (11)$$

when $\delta > 0$ (pushing into the L -wing), and $\rho = S/(S + |\delta|)$ when $\delta < 0$.

Slippage is:

$$\text{slippage} \equiv 1 - \rho \quad (12)$$

No other notion of slippage is needed.

The counter-policy received has sensitivity:

$$w_{\text{out}} = w_{\text{in}}^\rho \quad (13)$$

The residual (net unmatched sensitivity):

$$w_{\text{net}} = w_{\text{in}}^{1-\rho} \quad (14)$$

Properties of ρ .

1. **Marginal** ($|\delta| \rightarrow 0$): $\rho \rightarrow 1$. Infinitesimal commitments are matched at full efficiency.
2. **Finite trade**: $\rho < 1$. Always some slippage.
3. **Large effective size** ($|\delta| \rightarrow \infty$): $\rho \rightarrow 0$. Extreme commitments get almost nothing back.
4. **Deep pool** (K large, wing weights large): ρ stays near 1 even for moderate $|\delta|$.
5. **Skewed pool**: committing into the already-heavy wing yields lower ρ than committing into the light wing.

5.4 Deposit

The deposit (BTC-denominated cost extracted from the opposite wing) for a commitment with $\delta > 0$:

$$\text{deposit} = \frac{S \cdot |\delta|}{L + |\delta|} \quad (15)$$

For $\delta < 0$: $\text{deposit} = L \cdot |\delta|/(S + |\delta|)$. At balance and small $|\delta|$: $\text{deposit} \approx |\delta|$. At large $|\delta|$: deposit approaches the entire opposite wing weight. The opposite wing cannot be drained—a natural boundary.

6 State Evolution

Event	L	S	K	$P = S/L$
Commit into L -wing ($\delta > 0$)	$\uparrow$	$\downarrow$	$=$	$\downarrow$
Commit into S -wing ($\delta < 0$)	$\downarrow$	$\uparrow$	$=$	$\uparrow$
Add liquidity	$\uparrow$	$\uparrow$	$\uparrow$	$=$
Remove liquidity	$\downarrow$	$\downarrow$	$\downarrow$	$=$

Table 1: State evolution under market events. Trades move along the invariant curve (K preserved); liquidity changes shift the curve.

7 Unwinding and P&L

Commitments are live instruments, tradeable against the asset at any time. At entry, the trader committed a policy and received a counter-policy with efficiency ρ_{entry} . At exit, the trader returns the pair and receives asset at ρ_{exit} .

If the pool has moved toward the trader's view (consensus shifted toward φ_i): the commitment is closer to the trivial policy, and the trader unwinds at a *premium*. If the pool has moved against: the commitment is further from trivial, and the trader unwinds at a *discount*.

Total P&L has two components:

1. **Accumulated rebalancing P&L.** All deviations while held, filled at the locked-in ρ . The ongoing earned or paid spread on every rebalance.
2. **Unwind premium/discount.** The change in how the market values the commitment relative to the asset. Purely a function of how the pool state evolved.

8 Depth Profile and Hyperbolic Geometry

The equilibrium depth profile of the book decays exponentially from consensus:

$$D(\delta) \propto e^{-\alpha|\delta|} \quad (16)$$

where α is determined by pool depth (deeper pool $\Rightarrow$ slower decay). Price impact at deviation δ is:

$$\lambda(\delta) = \frac{\alpha}{D(\delta)} \quad (17)$$

growing exponentially toward the edges. Moving consensus is cheap near center (deep book), infinitely costly at the edges (no depth).

This depth profile defines a natural metric on deviation space:

$$ds = \frac{d\delta}{D(\delta)} = e^{\alpha|\delta|} d\delta \quad (18)$$

With the substitution $y = D(\delta)$:

$$ds^2 = \frac{dy^2}{y^2} \quad (19)$$

This is the Poincaré metric. The book's own depth profile *is* the hyperbolic geometry—not imposed, but derived from the self-consistent equilibrium shape.

The $\sinh$ in the invariant comes from integrating this exponential depth. Cumulative depth from consensus to deviation φ :

$$Q(\varphi) = \frac{D_0}{\alpha} \sinh(\varphi) \quad (20)$$

The invariant $L \cdot S = K$ is conservation of total resting commitment expressed in this natural coordinate.

9 Analogues in Traditional Finance

Perpetual swap. The sensitivity ratio ρ plays the role of the funding rate, but crystallized at entry rather than applied periodically. The reference path is trader-specific (their own drift) rather than spot price. There are no ongoing funding surprises—the full execution cost is embedded in the position at inception.

Cross-currency interest rate swap. Treating drift as an interest rate differential, the sensitivity ratio is the forward premium. Consensus φ^* is the implied forward rate. The $\sinh$ invariant is the shape of the forward curve, with exponential penalty for extreme rate views. A trader entering a policy is effectively swapping the market-implied rate for their own rate view, at a cost determined by the invariant.

10 Discussion

The mechanism unifies several functions that are typically separate: the expression of a directional view, the execution of that view as a rebalancing program, and the pricing of execution quality—all captured in a single primitive and priced by a single invariant.

The key structural innovation is the sensitivity ratio as permanent execution quality. In existing perpetual swap markets, the cost of maintaining directional exposure is repriced continuously through funding rates that the trader cannot predict at entry. Here, the execution quality is fixed at entry and applies uniformly to all future rebalances, transforming an unpredictable running cost into a known structural parameter.

The log-space convention for sensitivity is essential. Because w lives on a multiplicative scale with fixed point at 1, the natural interpolation between committed and trivial sensitivity is exponential: $w_{\text{out}} = w_{\text{in}}^\rho$ places the counter-policy at a well-defined fraction of the way (in log-space) between the committed policy and the trivial policy. The effective size formula $\delta = q \cdot \log(w) \cdot \sinh(\varphi - \varphi^*)$ inherits this convention, ensuring that the trivial policy ($w = 1$) produces zero effective size, and that trend-following ($w > 1$) and contrarian ($w < 1$) commitments at mirror sensitivities are mirror trades.

The consensus mechanism allows the entire market to be denominated in the underlying asset without external oracles. The asset is defined endogenously as the trivial policy at the market-clearing drift angle. Volatility—traditionally an exogenous parameter—is an emergent property of the disagreement among policy holders.

Open Questions

Several aspects remain for future work:

- The complex extension of the formalism, if motivated by insufficiency of the real-valued framework.
- The fractal/multi-scale structure of rebalancing across timescales.

- The discrete order book implementation as computational substrate.
 - Margin and collateral mechanics.
 - The angle-to-velocity map $v(\varphi)$ and its joint derivation with the pricing function.
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Notation Summary

Symbol	Meaning
(φ, w, q)	Policy: drift angle, sensitivity, units
φ^*	Consensus drift (heading)
L, S	Wing weights: $\sinh(\Phi_L), \sinh(\Phi_S)$
$K = L \cdot S$	Invariant (conserved across trades)
$\text{tilt} = (L - S)/(L + S)$	Pool asymmetry
$P = S/L$	Marginal quote
$\delta = q \cdot \log(w) \cdot \sinh(\varphi - \varphi^*)$	Effective size (signed)
$\rho = L/(L + \delta)$	Sensitivity ratio (value-preservation efficiency)
$1 - \rho$	Slippage
$w_{\text{out}} = w_{\text{in}}^\rho$	Counter-policy sensitivity
$w_{\text{net}} = w_{\text{in}}^{1-\rho}$	Residual (unmatched) sensitivity
$D(\delta) \propto e^{-\alpha \delta }$	Equilibrium depth profile